

MLOps Pipeline for Bird Species Identification from Audio

Group 2:

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Machine Learning and Data in Operation

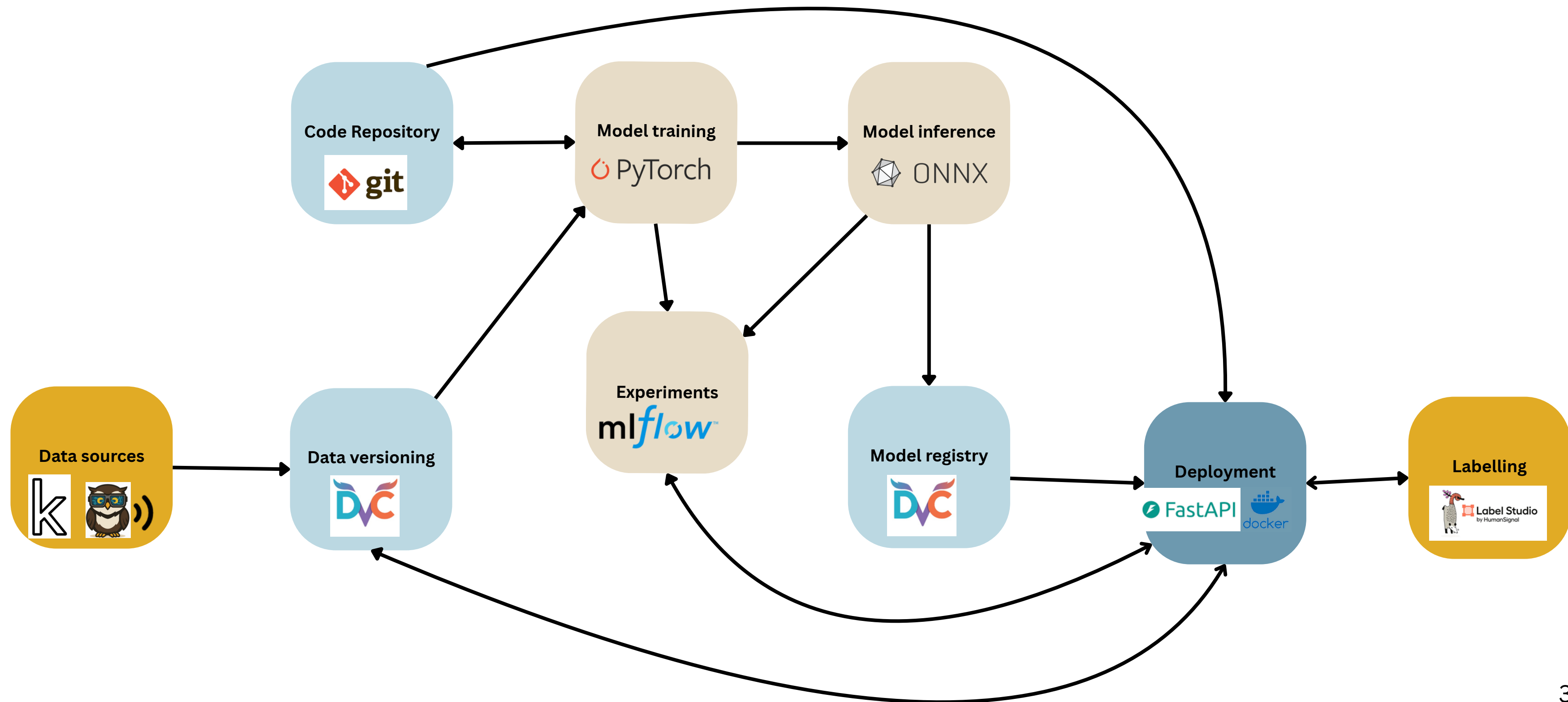
05.06.2025



Project Overview

- **Objective:** Develop and deploy machine learning models to classify animal species from audio data, recorded in Colombia's Magdalena Valley, supporting ecological conservation
- **Challenges:** Address limited labeled data and weak labeling by building robust pipelines
- **Approach:** Datasets versioned with DVC, containerized solution with Docker, prediction endpoints using FastAPI, Label Studio setup to enable expert and model labeling, enriching the dataset over time.

Project Flow



Data Versioning & Tracking



Reproducibility and collaboration: DVC tracks versions of models and datasets allowing sharing and reproducibility.



Storage Efficiency: Large audio datasets are efficiently stored remotely on an Azure server.



SSH Access: secure and authenticated access to the data and model repository

Fully Modular Training



 **Audio Preprocessing** Sample Rate, Segment Length, Mel Settings ...

 **Training Setup** Backbone, Training Strategy, Hyper-Parameters ...

 **Experiment Tracking** Name, Tags, Metrics, Model Artifacts ...

All parameters **selected via one config file**

Model Conversion



Load Trained Model

- Flexible, config-driven process.

Convert to ONNX

- Automated pipeline setup.

Validate Model

- Automatic verification of model correctness.

Inference Pipeline



Inference Factory

- SOLID-driven **extendable modular** pipeline code

Predictions

- Inference pipeline optimized for speed (**below 0.5** seconds on CPU).

Results

- **Top-k** species predictions, **confidence** thresholds

Currently **top 100** on Kaggle.

← →

MLOWLS_Training [SSH: ab19]

03

! config.yaml X

! config.yaml

18 training:

21 learning_rate: 1e-3

22 weight_decay: 1e-4

23 val_fraction: 0.1 # - hold out 10% of train.csv for validation

24 training_strategy: "cutmix" # or "standard" or "mixup"

25 mixup_alpha: 0.4 # for mixup strategy

26 cutmix_alpha: 1.0 # for cutmix strategy

27 early_stopping_patience: 10

28 accuracy_top_k: 1 # for top-k accuracy

29 auc_average: "macro" # AUC averaging strategy

30

31

32 model:

33 backbone: efficientnet_b0

34 num_classes: 206

35 dropout: 0.3

36

37 experiment:

38 name: birdclef25_robust_train

39 tags:

OUTPUT

TERMINAL

PROBLEMS

DEBUG CONSOLE

PORTS 1

bash + ~

o (project) christian.pala@19-host:~/MLOWLS_Training\$ mlowls-train --config config.yaml

SSH: ab19

6a4c6dd7 ↺

0 0 0

1

Git Graph

christian_pala (6 days ago)

Ln 34, Col 1 (28 selected)

Spaces: 2

UTF-8

LF

{ } YAML

API & Deployment

FastAPI

-  **Production-Ready Performance:** Lightweight, high-speed async endpoint for real-time inference
-  **Health checks:** Liveness & Readiness endpoints for monitoring and orchestration
-  **Automated integration with Label Studio:** model predictions automatically uploaded on Label Studio

BirdCLEF Inference API

1.0.0 OAS 3.1

openapi.json

BirdCLEF 2025 Inference API

default

GET	/	Read Root	
GET	/liveness	Liveness Check	 
GET	/readiness	Readiness Check	
POST	/predict	Predict Endpoint	

Schemas

Body_predict_endpoint_predict_post	Expand all	object
HTTPValidationError	Expand all	object

API & Deployment



-  **All-in-One MLOps Stack:** API + MLflow + Label Studio + DVC Pull + DVC Sync in single automated deployment container
-  **Smart Service Orchestration:** Supervisor manages 5 services with dependency priorities
-  **Scalability Ready:** Single 'docker-compose up' command for entire ML infrastructure
-  **Automated Data Pipeline:** Automated data pull/push services ensures near real-time remote synchronization

```

1  # Dockerfile
2  FROM python:3.12-slim
3
4  # Set environment variables
5  ENV PYTHONDONTWRITEBYTECODE=1 \
6      PYTHONUNBUFFERED=1 \
7      PIP_NO_CACHE_DIR=1 \
8      PIP_DISABLE_PIP_VERSION_CHECK=1 \
9      LABEL_STUDIO_LOCAL_FILES_SERVING_ENABLED=true \
10     LABEL_STUDIO_LOCAL_FILES_DOCUMENT_ROOT=/app \
11     LABELSTUDIO_TOKEN="insert_your_token_here"
12
13 # Install system dependencies
14 RUN apt-get update \
15     && apt-get install -y --no-install-recommends \
16         supervisor \
17         git \
18         build-essential \
19         ffmpeg \
20         libsndfile1 \
21         curl \
22         openssh-client \
23     && rm -rf /var/lib/apt/lists/*
24
25 # Create user and group early
26 RUN groupadd -r appuser && useradd -r -g appuser appuser
27
28 # Set working directory and change ownership to appuser
29 WORKDIR /app
30 RUN chown appuser:appuser /app
31
32 # Clone repository as appuser
33 USER appuser
34 RUN git clone --depth=1 \
35     --branch main \
36     https://github.com/Manuel-Ippolito/MLOWLS_Deployment.git /app
37
38 # Switch back to root to create system directories
39 USER root
40
41 # Create directories that need root permissions and set ownership
42 RUN mkdir -p /app/data /app/mlflow/mlruns /opt/venv /home/appuser/.ssh \
43     && chown -R appuser:appuser /app/data /app/mlflow /opt/venv /home/appuser
44
45 # Switch back to appuser for remaining operations
46 USER appuser

```

Labeling



-
- **Human-in-the-loop** feedback embedded in the design of the pipeline, to improve data quality and model accuracy
 - **API integration** between Label Studio and our API, to incentivize and simplify data collection efforts
 - **Automated** versioning and storing of the newly labeled audio files, to be ready for the next improvement cycle

Labeling - Demo



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- Easy-to-use interface to navigate through data (listening)
 - Data Cleaning
 - Ground Truth
 - Annotation from different experts
 - Evaluation from different models as reference

Codebase

- https://github.com/Manuel-Ippolito/MLOWLS_Deployment
- Contact us for sensitive information needed to run the code.

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Q&A

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